

Model Solution with Fixed Prices

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Price name: $p^n(x) = p > 0$

parameter = fixed price

Model solution w/ fixed price

Need to find tightness x , which is given by.

$$y^d(x, p) = y^s(x) \leftarrow \text{AD=AS implicitly defines } x$$

$$\frac{x^z}{[1 + \tau(x)]^{z-1}} \cdot \frac{\mu}{p} = f(x) \cdot k$$

x^z : utility parameters
 $[1 + \tau(x)]^{z-1}$: matching wedge
 μ : fixed price
 p : price
 $f(x)$: selling proba.
 k : aggregate capacity
 both in RA & HA model

AD curve
(pure aggregate demand)

AS curve

Rewrite tightness equation.

$$\frac{x^z}{[1 + \tau(x)]^{z-1}} \cdot \frac{\mu}{p} = f(x) \cdot k \quad \lambda(x) = 0$$

• $x = 0, \quad f(x) = 0 \quad \tau(x) = \frac{p}{1-p}$

$$\lambda(0) = \frac{x^z}{(p/(1-p))^{z-1}} \cdot \frac{\mu}{p} > 0$$

• $x = x^m \quad \tau(x) = \infty \quad f(x^m) > 0$
 $\lambda(x^m) = - \frac{f(x^m)}{f'(x^m)} k < 0$

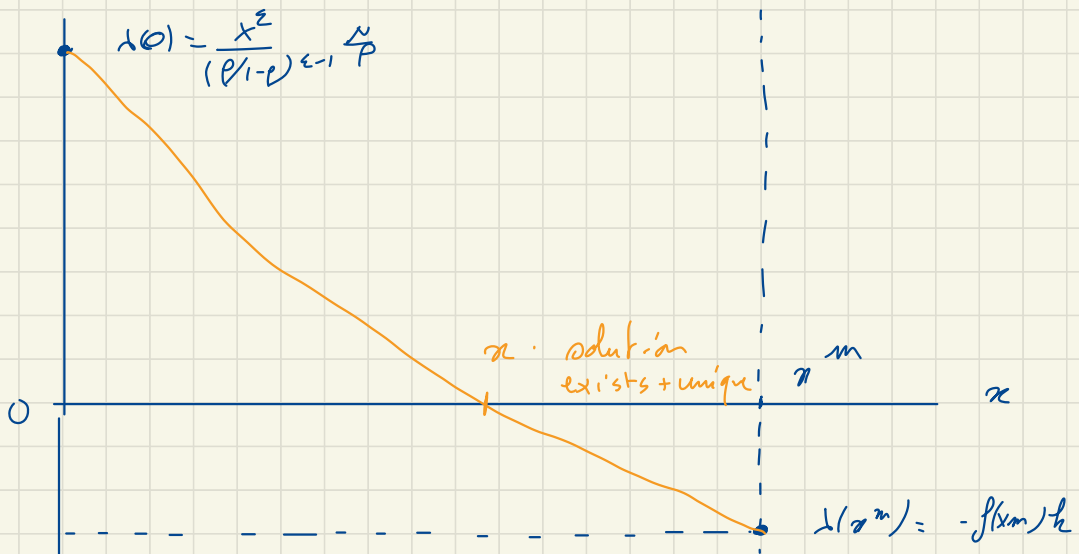
- $\lambda(x)$ is continuous

Intermediate value thm. there is x such that $\lambda(x) = 0 \rightarrow$ our model has (at least) one solution.

- $\lambda(x)$ is strictly decreasing | $\tau(x)$ is strictly increasing, $\varepsilon > 1$, $f(x)$ is strictly increasing)

$\rightarrow x$ such that $\lambda(x) = 0$ is unique $\rightarrow$ our model solution is unique.

$\rightarrow$ model has always unique solution



Another representation:

